

T E R

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M1 Actuariat
ISFA



IMPACT OF EXCESS-OF-LOSS
REINSURANCE ON CATASTROPHE
RISK IN NON-LIFE INSURANCE

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1 Abstract

1.1 Abstract

This project investigates the impact of excess-of-loss reinsurance on the loss distribution of a non-life insurance portfolio exposed to catastrophic risks. First, aggregate losses are modeled using a frequency-severity framework, combining a Poisson distribution for claim frequency with a heavy-tailed distribution for claim severity.

Monte Carlo simulations are then performed to estimate the annual loss distribution and assess risk using standard measures such as Value-at-Risk (*VaR*) and Tail Value-at-Risk (*TVaR*). In a second step, an excess-of-loss reinsurance treaty is introduced, and its effect on the loss distribution is analyzed.

The results show that reinsurance significantly reduces extreme losses and mitigates tail risk, with a particularly strong impact on TVaR. This study highlights the key role of reinsurance in managing catastrophic risk exposure for insurance companies.

Following sections items are clickable

1.2 Forewords

We refer to results and demonstrations written in auxiliary papers. In a first time if the github link doesn't show the paper, just refresh the page.

If the github link still doesn't work, you can contact us in a second time at : mhreinette@gmail.com. A other version of this paper is available in a light theme on this present link.

All the resources and notebook are available on this [github repository](#). Different colors are used to characterise : [definitions](#), [theorem](#), [important results](#), [critics](#).

1.3 Updates

Success notes on the recent updates of this paper :

- Finished abstract
- Finished definition/notations environment
- Finished theorem environment

1.4 tests

Toutes les ressources utilisées seront présentes en annexes et sur le *repository github* ci-dessus.

2 Annex

2.1 Annex

2.2 Notations Index

◦ σ -Algebra : $\mathcal{A}$

Let : $\mathcal{A}$, define by :

- Ω : a space
- $\Omega \in \mathcal{A}$
- $\forall A \in \mathcal{A}, \bar{A} \in \mathcal{A}$
- $\forall (A_n)_{n \in \mathbb{N}} \in \mathcal{A}^{\mathbb{N}}, \bigcup_{n \in \mathbb{N}} A_n \in \mathcal{A}$
 - Let « the borelian σ -algrebra » : $\mathbb{B}(\mathbb{R}^n)$ & λ « Lebesgue mass function ».

definitions and notations

◦ Probabilised Space : $(\Omega, \mathcal{A}, \mathbb{P})$

Let : $(\Omega, \mathcal{A}, \mathbb{P})$, define by :

- Ω : a space
- $\mathcal{A}$: a σ -algebra of Ω
- $\mathbb{P}$: a probability mass function

definitions and notations

◦ Filtered Space : $(\Omega, \mathcal{A}, \mathcal{F} = (\mathcal{F}_t)_{t \in T}, \mathbb{P})$

Let : $(\Omega, \mathcal{A}, \mathcal{F} = (\mathcal{F}_t)_{t \in T}, \mathbb{P})$, define by :

- $(\Omega, \mathcal{A}, \mathbb{P})$: a probabilised space
- $\mathcal{F} = (\mathcal{F}_t)_{t \in T}$: the natural filtration of a Brownien motion W

definitions and notations

◊ Theorem

Hypothesis :

- test
- test

Conclusions :

- conclusion

theorem

2.3 Bibliography